
SYMBOLIC REGRESSION OF GRAVITATIONAL WAVES PARAMETERS

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ABSTRACT

The problem addressed in this work is the creation of a model for gravitational wave detection, achieved with minimal intervention in the data itself, such as signal filtration and scaling using constants related to the wave detector. This model was further utilized for data upon which symbolic regression was subsequently performed, consisting of finding parameters A_i and ω_i from the equation $y_i(A_i, \omega_i; t) = A_i \cdot \sin(\omega_i t)$, where A_i denotes amplitude, ω_i denotes frequency, and t denotes time. The data from the symbolic regression are then utilized together with the model's result for data analysis. Unfortunately, the graphs do not show any dependency between detection probability and frequency or amplitude; further possibilities for work are outlined in the conclusion.

1 Introduction

Gravitational waves are disturbances in spacetime that propagate at the speed of light and are generated by the motion of very massive and accelerated objects, such as pairs of black holes or neutron stars. Their existence was predicted by Albert Einstein within the framework of the General Theory of Relativity as early as 1916.

The direct detection of gravitational waves was only achieved in 2015 by the LIGO experiment, which recorded a signal from the merger of two black holes. This discovery confirmed the validity of the General Theory of Relativity under extreme conditions and opened a new field of observing the universe: gravitational wave astronomy. Today, gravitational waves represent an important tool for studying compact objects and the dynamics of strong gravitational fields. The first gravitational waves were detected using other gravitational waves; however, over time, new methods are being created, for example, based on statistics. This work deals with detection using a neural network and an attempt to understand their physical basis using symbolic regression, with the help of parameters A_i and ω_i of the function $y_i(A_i, \omega_i; t) = A_i \cdot \sin(\omega_i t)$.

Keywords Gravitational waves · Symbolic regression · Neural network

2 Detection

The work can be divided into 2 parts: the first is the creation of the model itself, and the second is the application of symbolic regression. The result is then combined into a data analysis. The creation of the model itself includes dataset analysis, data preparation for model training, finding a suitable model for the specific situation, and finally training and final validation.

Loading the data was problematic because 560,000 files, each with 3 time series from 3 detectors (LIGO Hanford, LIGO Livingston, Virgo), consisting of 4096 numbers in the order of $1e - 20$, hardly fit into memory. Therefore, data loading is done using a dictionary (file:result), where the actual file loading is performed only when needed. This further slows down the learning itself, which can cause slow learning on weaker computers. Therefore, I decided to use a faster model for data fine-tuning, but use a more complex model for the final model.

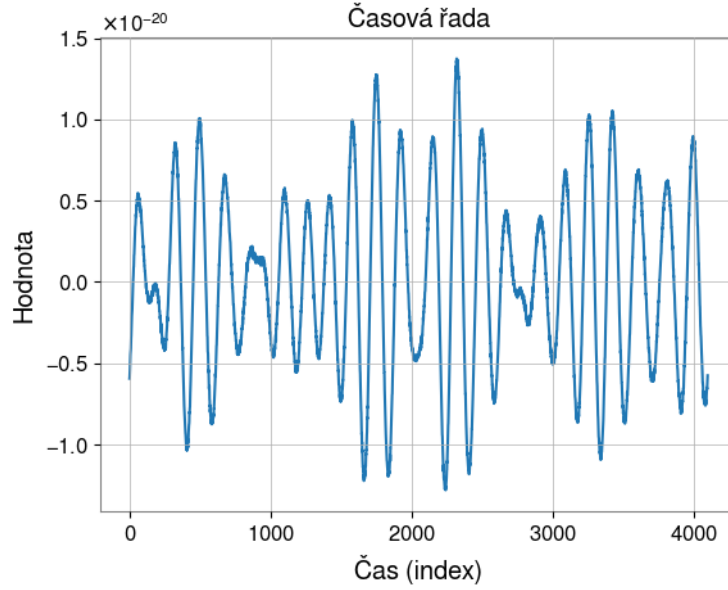


Figure 1: Unmodified time series, Y axis are values, X axis is time(index)

For the model to be successful at all, the data needs to be processed. The dataset contains 560,000 .npy files, each containing 3 time series. Each is from a different specific detector. In their basic form, the data take on very small values and are full of noise. There are several options that can be applied to the data to solve these problems:

- **Scaling** – Data are brought into values that computers can realistically work with. The first and second detectors are scaled by $1.5e - 20$, and the third detector is scaled by $0.5e - 20$.
- **Filtering** – Data contain a lot of noise or other signals that are not the gravitational waves themselves; filtering them out creates a cleaner signal.
- **Tukey window** – The Tukey window smooths the edges of the signal so that artificial frequencies do not arise during spectrum calculation. This reduces spectral leakage and improves frequency analysis accuracy. Spectral leakage is a phenomenon where the energy of one frequency spills over into surrounding frequencies because the analyzed signal is time-limited and has sharp edges. False or blurred frequencies that are not in the actual signal then appear in the spectrum.

I personally spent a considerable amount of time on which of these options to use. Ultimately, I trained the model on data that was modified as little as possible, while ensuring the model still worked. I, therefore, used scaling and filtering using a Butterworth filter. This is a type of digital or analog filter that has the smoothest possible frequency response without ripple in the passband. It is used to smoothly suppress unwanted frequencies (e.g., noise) without significant distortion of the signal amplitude. Thus, unwanted frequencies are filtered out of the signal, and at the same time, it is scaled to values that computers can work with, preventing the entire calculation from being discarded due to numerical errors.

To refine how much to process the data, I used a small 1D CNN model. The model from this network accepts an input with three channels and consists of four consecutive blocks. Each block contains a 1D convolution, followed by batch normalization and an activation function. The network gradually reduces the size of the convolution kernels ($k=64,32,16,8$). Dimensionality reduction is ensured by MaxPool1d layers after the first three blocks. The final block uses global averaging to create a feature vector, which is subsequently processed by a fully connected linear layer into a single output neuron for classification. The advantage of this simpler network is the ability to test different data processing methods because learning on this network takes hours, instead of days as with the final network.

The result of this step was the previously mentioned use of only 2 steps for data processing. The reason was to use less processing so that the data for both detection and symbolic regression would be usable in the same state. Using a Tukey window would deform the wave itself, so it was not used for detection.

For the final model, a network based on the same basis but more complex was used; during learning, various augmentation techniques were used, such as shifting and swapping inputs. The purpose of this is to prevent the

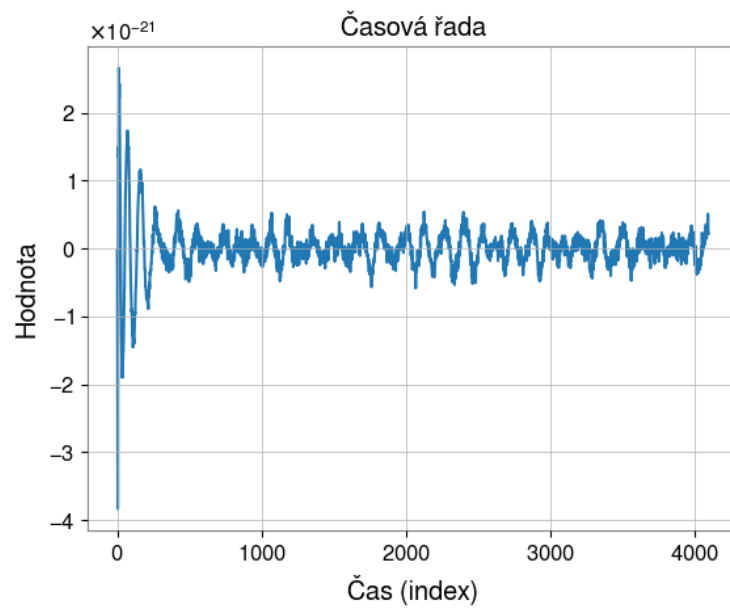


Figure 2: Series after applying Butterworth filter, Y axis are values, X axis is time(index)

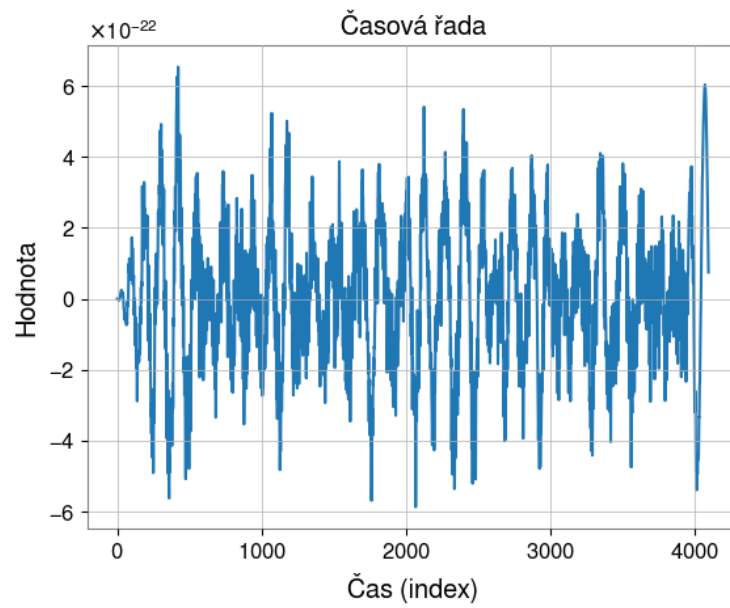


Figure 3: Series after applying Butterworth filter and Tukey window, Y axis are values, X axis is time(index)

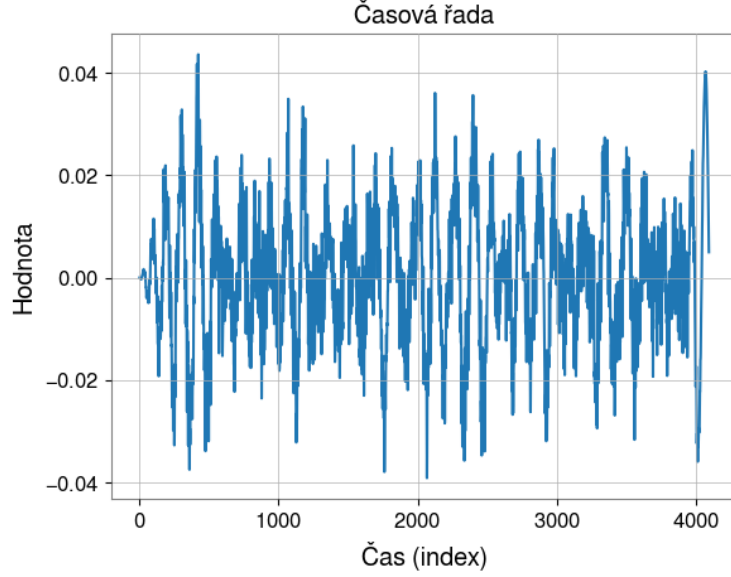


Figure 4: Series after applying Butterworth filter, Tukey window and scaling, Y axis are values, X axis is time(index)

model from learning on only one input and potentially to prevent learning on bad data caused by the use of filters. The final model was then further tested on test data, where the model achieved $ROC - AUC = 0.8398$, which differs from the best results in the Kaggle competition only by hundredths, considering the lesser processing, I viewed this as a good result.

3 Symbolic Regression

This section focuses on symbolic regression. Since the equation for which I want to determine the parameters is known in advance, it is not necessary to use symbolic regression directly to determine parameters iteratively, for example using a genetic algorithm. I tested this, but given the huge space of possible solutions, it was slow and the results were visually poor. This solution would be suitable if I had to find some function, or if simpler methods did not exist, such as curve fit, which uses non-linear least squares for fitting function f to data.

Here, however, one modification is important, and that is the addition of the parameter ϕ_i . This parameter serves as a phase shift because not every wave starts exactly at 0—indeed, hardly any starts exactly at 0. This parameter is only important for better and more accurate results; returning it provides nothing further for data analysis. The problem is therefore to find suitable A_i and ω_i for the equation:

$$y_i(A_i, \omega_i; t) = A_i \cdot \sin(\omega_i t + \phi_i)$$

Another problem is finding suitable initial values for A_i and ω_i . While for A_i this parameter is something between 1 and 0, which is not such a large range and `curve_fit` is able to find a good approximation at $A_1 = 1$, for ω_i it is a bigger problem; the initial estimate greatly influences the final result. Therefore, it is necessary to find an initial estimate of ω_i using a method. For this, I used FFT (Fast Fourier Transform), which is a fast algorithm that converts a signal from the time domain to the frequency domain and allows determining which frequencies are contained in the signal. This ensures a good initial estimate for ω_i , where `curve_fit` takes care of the rest.

The result of `curve_fit`, or symbolic regression, is therefore A_i , ω_i , the variance of estimates, and the RMSE of estimates can be calculated. I then pack the data from detection and symbolic regression into a dataset and save it in csv format. In this way, I prepared 3 datasets, differing in how much the data are processed for symbolic regression:

- **dataset_1** – For the first dataset, I used the same data processing as for detection; therefore, the best dependency between the model result and the symbolic regression result is there.
- **dataset_2** – The second dataset contains data that are only rescaled.
- **dataset_3** – Compared to the first dataset, this has data to which I also applied Tukey window.

In most cases, dataset_1 is used unless explicitly stated otherwise. The reason for not using other datasets in the graphs is that nothing new appeared in the data, so it is pointless to show these graphs.

4 Results

4.1 Detection

The result of the model on new data is:

- Validation Accuracy: 0.7769
- F1 score: 0.7269
- ROC-AUC: 0.8398

For further results, I again used data that the model did not see during training; the same data were used for both detection and symbolic regression. First, however, the detection results:

Table 1: Distribution of predictions across wave detection probability.

Prob Range	Total	True Label	False Label
0.0 – 0.2	15531	3180	12351
0.2 – 0.3	12949	3853	9096
0.3 – 0.4	6160	2486	3674
0.4 – 0.5	3096	1571	1525
0.5 – 0.6	1862	1154	708
0.6 – 0.7	1275	944	331
0.7 – 0.8	1058	852	206
0.8 – 0.9	1121	1002	119
0.9 – 1.0	12948	12906	42

According to these results, it could be said that the model itself has problems detecting in the range 0.4-0.6, where the result is almost random. This may mean that the model itself has problems, or that the data are not good enough. Since these are borderline measurements and the model determines most data correctly, it can be concluded that these are weaker gravitational waves containing a lot of noise that may not have been filtered out. However, more data where we know the label would be needed for this, given that the data are borderline.

4.2 Symbolic Regression

The results of the regression itself are graphs of the dependence of amplitude on frequency, or the empirical distribution of normalized data. These data are generated from the symbolic regression itself, so the output of the detection model is not needed for them. The displayed data are from only one detector, but the result did not change between them; due to the scope of the file itself, I did not include these graphs in the work.

From these graphs, it is visible that the data indeed have dependencies regarding frequency and amplitude, and there is certainly some dependency on detection, since without applying the Butterworth filter the model could not detect (it chose randomly). The values are more around a single value; this is more visible with frequency. The question, however, is whether this is an imperfection of the model for symbolic regression (a function with constant frequency and amplitude cannot describe gravitational waves well), or a lack of data for other frequencies and amplitudes. What is important to say, however, is that whether the wave itself is noise or truly a gravitational wave does not influence the dependence of frequency and amplitude; both groups occur in the same places.

4.3 Symbolic Regression with Detection

In this section, the detection results are combined with the symbolic regression result.

The graphs of the dependencies of normalized amplitude on probability and normalized frequency on probability show a created shape like a "lying vase". However, this shape is likely (the work contains a graph of the estimate of the mean value and variance from the data, which is homoscedastic over the entire series) due to a lack of data for the given detection probability, rather than any dependencies.

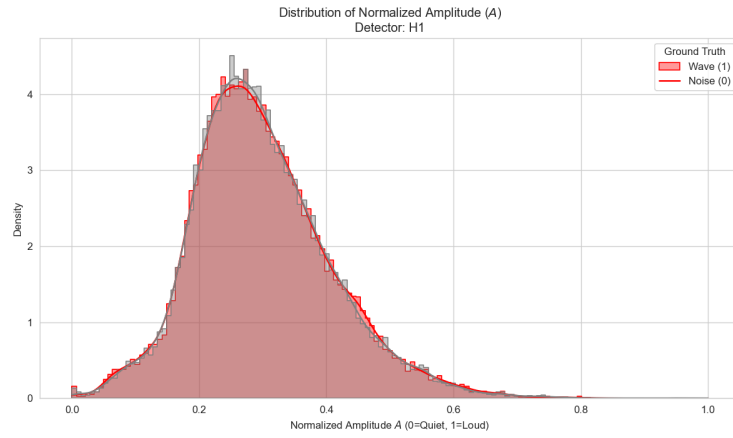


Figure 5: Empirical distribution of normalized amplitude of the first dataset (Butterworth and scaling applied)

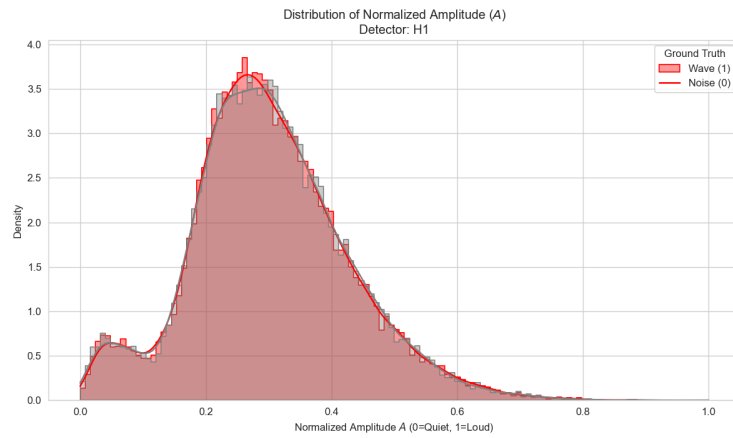


Figure 6: Empirical distribution of normalized amplitude of the second dataset (scaling), likely unremoved noise is visible compared to dataset_1

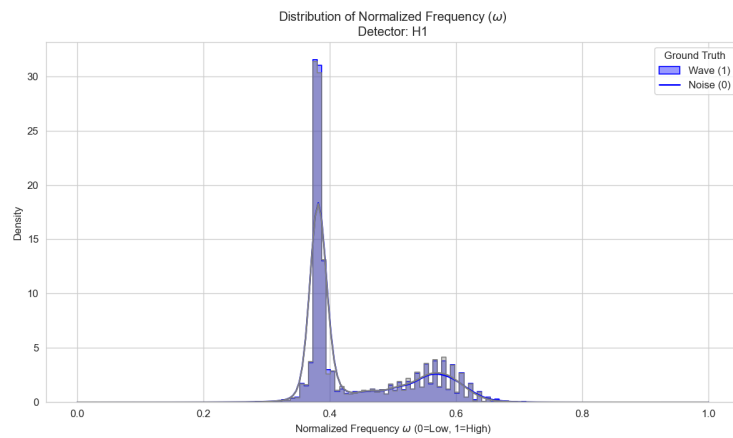


Figure 7: Empirical distribution of normalized frequency of the first dataset (Butterworth and scaling applied)

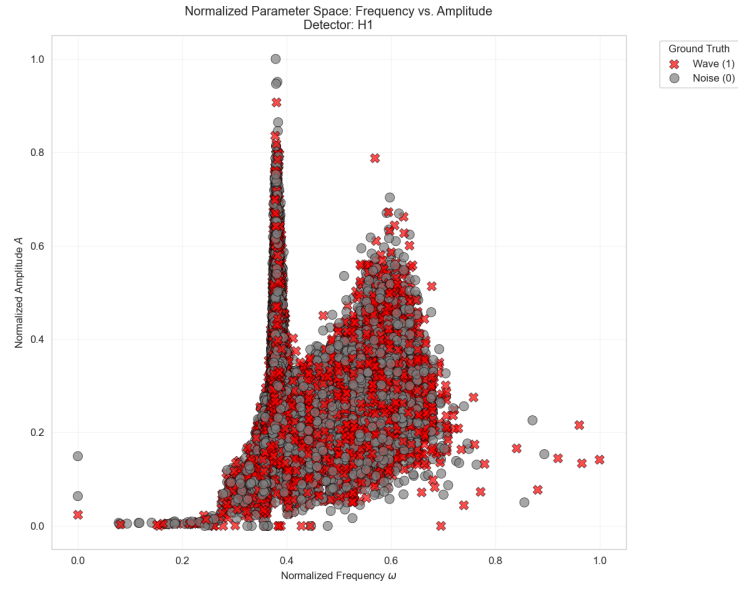


Figure 8: Graph of dependence of normalized amplitude on normalized frequency of the first dataset

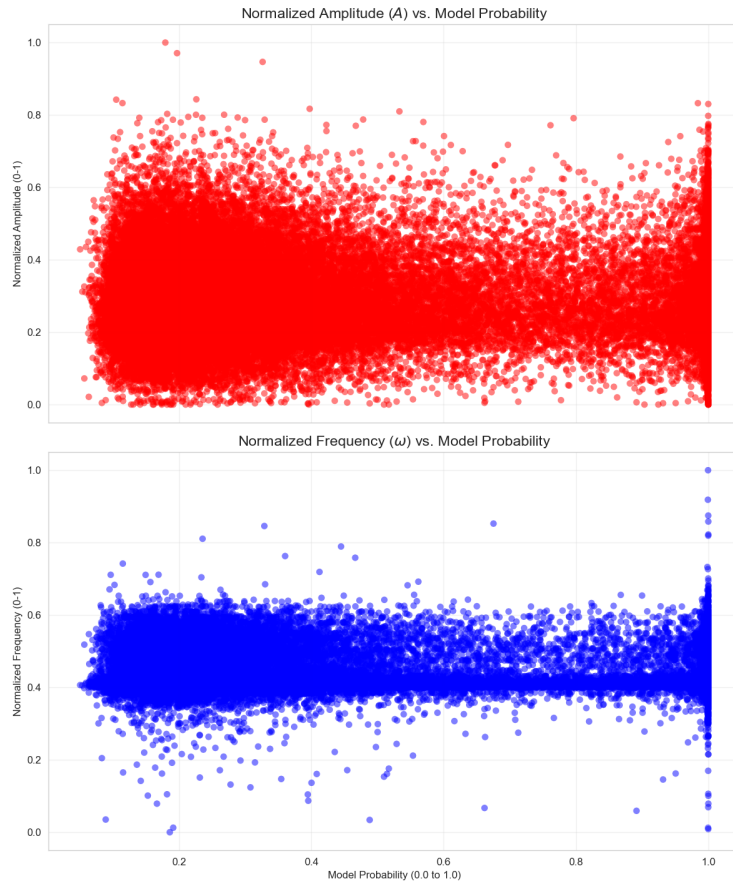


Figure 9: Graphs of normalized amplitude and normalized frequency against detection probability

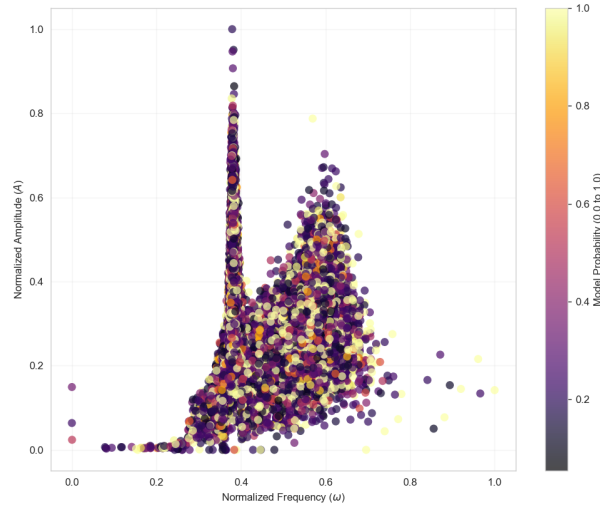


Figure 10: Graph of dependence of normalized amplitude on normalized frequency of the first dataset with a color scheme according to detection probability

The graph of amplitude versus frequency was further extended with a color scheme representing the probability that the wave is a gravitational wave. As with symbolic regression itself, no dependence of amplitude or frequency on detection probability is visible here.

Estimate of mean value and variance. It is the same for all datasets; the estimates are homoscedastic and have the same value.

The last graph is then the measured error vs probability, the same shape as with amplitude and frequency, again likely due to a lack of data for the given probability.

5 Conclusion

Despite the fact that the work successfully includes both detection and finding parameters A_i and ω_i , the conclusion of the work is negative. Personally, I did not find a dependence of detection on amplitude or frequency in the data. Estimates of mean values and variances confirm this conclusion. For full confirmation, a lot of new data would be needed; the competition itself is several years old, and a lot of research regarding gravitational waves has taken place during that time.

If even new data did not prove a dependence, there is still the possibility of a better equation for explaining gravitational waves. The problem, however, is in the output of these functions. In the case of this work, there are only 2; other functions may have many parameters that are then difficult to interpret. It would, therefore, be necessary to find a balance between the function's ability to explain gravitational waves and its interpretation for subsequent analysis.

6 Links

6.1 Data

<https://www.kaggle.com/competitions/g2net-gravitational-wave-detection/data>

6.2 Code

<https://gitlab.fit.cvut.cz/pokorlu8/mvi-sp>

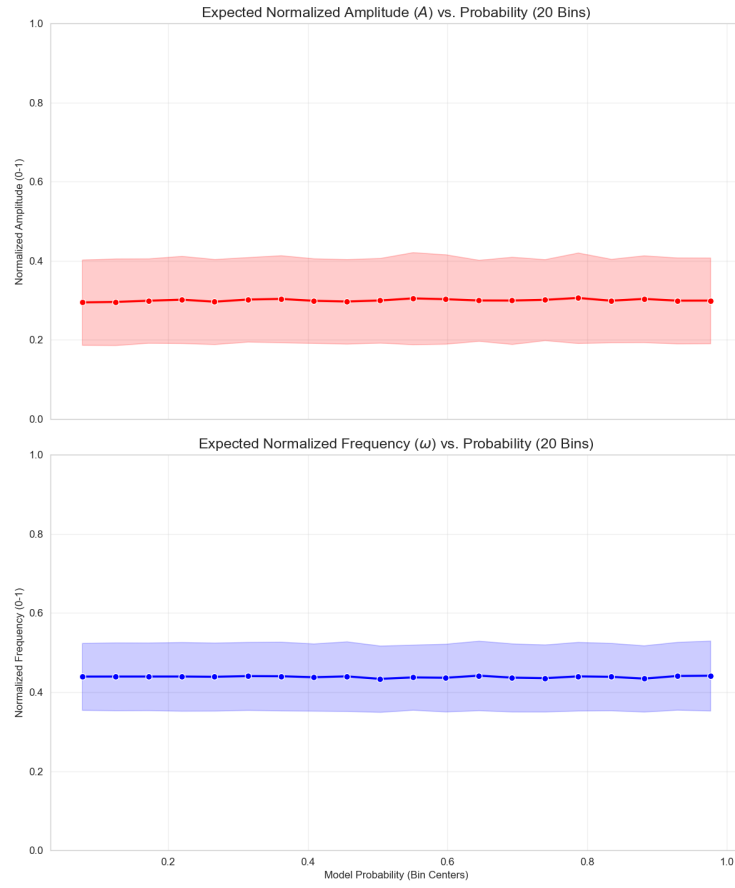


Figure 11: Estimate of mean values and variance of normalized amplitude and frequency with dependence on detection probability

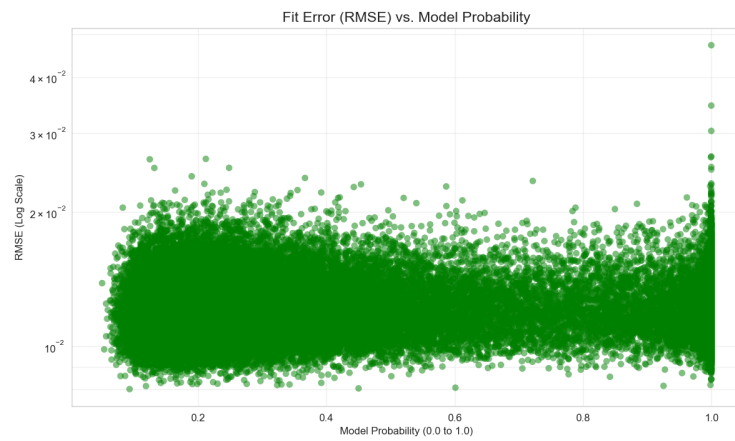


Figure 12: Dependence of RMSE on detection probability

6.3 Sources

<https://www.kaggle.com/competitions/g2net-gravitational-wave-detection/leaderboard>

<https://www.kaggle.com/competitions/g2net-gravitational-wave-detection/writeups/kdl-top-1-solution-deep-learning-part>

<https://www.kaggle.com/code/firefliesqn/g2net-gpu-newbie>

<https://iopscience.iop.org/article/10.1088/1361-6382/ab685e>

<https://gwpv.github.io/docs/3.0.12/>

<https://www.kaggle.com/competitions/g2net-gravitational-wave-detection>